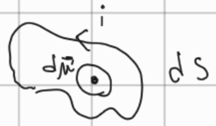
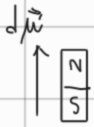
MOMENTO ANGOLARE INTRINSECO : SPIN

Princ. Equiv. Ampere :

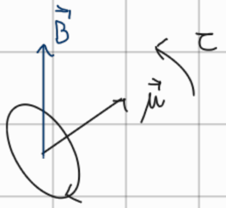


$\Longleftrightarrow$



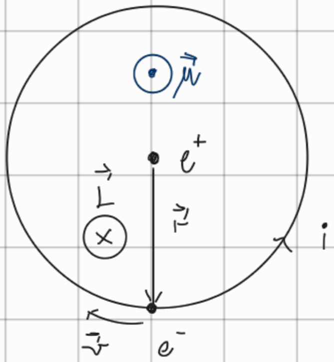
$d\vec{\mu} = i\,dS\,\vec{u}_n$

$\Rightarrow$



$\vec{\tau} = \vec{\mu} \wedge \vec{B}$, $U_{Dip.} = -\vec{\mu}\cdot\vec{B}$

$\rightarrow$ FISICA ATOMICA:



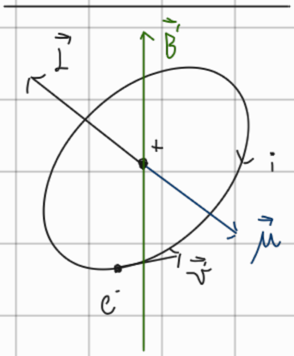
$\vec{\mu} = i\,S\,\vec{u}_n$

$i = \frac{e}{T} = \frac{e}{\frac{2\pi r}{v}} = \frac{e v}{2\pi r}$

$\vec{\mu} = \frac{e v}{2\pi r}\pi r^2\,\vec{u}_n = \frac{e v}{2} r\,\vec{u}_n =$

$= \frac{e}{2m}\overbrace{m v r}^{|\vec{L}|}\,\vec{u}_n = \boxed{-\frac{e}{2m}\vec{L} = \vec{\mu}}$

Nella fisica delle part. $\vec{L} \Leftrightarrow \vec{\mu}$, $\boxed{\gamma = -\frac{e}{2m}}$ $\Rightarrow$ RAPPORTO GIROMAGNETICO



- $\vec{\tau} = \vec{\mu}\wedge\vec{B} = -\frac{e}{2m}\vec{L}\wedge\vec{B} = \frac{e}{2m}\vec{B}\wedge L$

- $\frac{d\vec{L}}{dt} = \vec{\tau}$ (MOM. TORCENTE) $= \frac{e}{2m}\vec{B}\wedge\vec{L}$

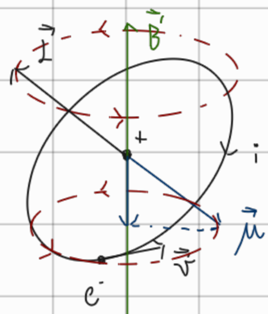
$\frac{d\vec{L}}{dt} = \frac{dL}{dt}\hat{u}_L + \vec{\omega}_L\wedge\vec{L} = \frac{e}{2m}\vec{B}\wedge\vec{L}$

a. $|\vec{L}| = \text{cost}$

b. $\frac{d}{dt}(\vec{L}\cdot\vec{B}) = 0 \Rightarrow \measuredangle\ \vec{L}\text{ e }\vec{B}$ RESTA COSTANTE

c. $\vec{L}$ RUOTA ATTORNO a $\vec{B} \Rightarrow |\vec{\omega}_L| = \frac{e|\vec{B}|}{2m}$

$\Rightarrow$ PRECESSIONE di LARMOR



DIAMAGNETISMO

$$\omega_z = \frac{eB}{2m} \cong 10^{11} \div 10^{12} \ rad/s \ll \omega_0 = 10^{16} \ rad/s$$

($B \sim 10\,T$: rivoluzionale / tipicamente; ω_0: rag. Bohr)

(precessione lenta e trascurabile)

CAMPO MAG. non unif.: $\vec{F} = -\nabla U_{dip} = -\nabla(-\vec{\mu}\cdot\vec{B}) = \{\vec{B} = \vec{B}(z)\,\vec{u}_z\} \Rightarrow$

$$\vec{F} = \mu_z \frac{\partial B_z}{\partial z}\vec{u}_z$$

UNITÀ NATURALE $\Rightarrow \hbar$ [mom. ang.]

$$\vec{\mu} = -\frac{e}{2m}\vec{L} = -\underbrace{\frac{e\hbar}{2m}}_{\text{MAGNETONE di BOHR}}\frac{\vec{L}}{\hbar} = -\underbrace{g_\ell}_{\text{FATTORE g-ORBITALE}}\mu_B\frac{\vec{L}}{\hbar}$$

MAGNETONE di BOHR $\mu_B = \frac{e\hbar}{2m}$

FATTORE g-ORBITALE $g_\ell = 1$

$$\mu_B = 9{,}3 \cdot 10^{-24}\ J/T = 5{,}8\cdot 10^{-5}\ eV/T$$

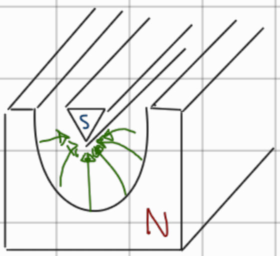
$$\Rightarrow [L^2 = \hbar^2\ell(\ell+1)] \Rightarrow \boxed{|\vec{\mu}| = \mu_B\sqrt{\ell(\ell+1)}}$$

$$L_z = \hbar m \Rightarrow \boxed{\mu_z = -g_\ell\,\mu_B\frac{L_z}{\hbar} = -\mu_B m}$$

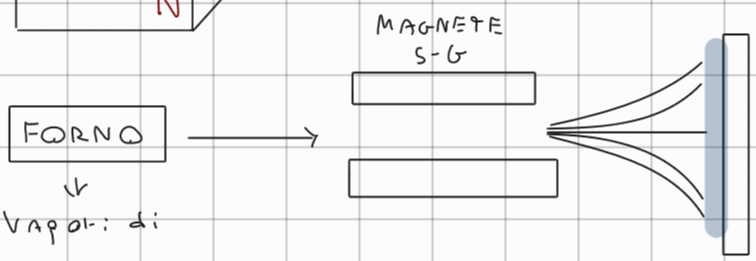
$$\Rightarrow \vec{F} = \mu_z\frac{\partial B_z}{\partial z}\vec{u}_z = -\mu_B m\frac{\partial B_z}{\partial z}\vec{u}_z$$

se il campo mag. non unif. F é in funz. di m

[1923] Stern - Gerlach



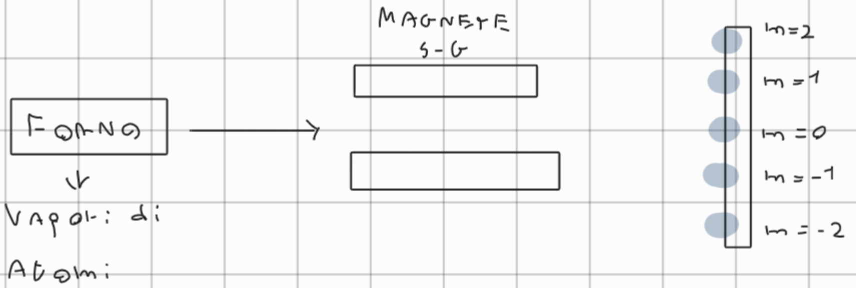
$$\frac{\partial B_z}{\partial z} = 1000\ T/m$$ (molto più forte verso la punta)



Vapori di Atomi

① Fisica Classica

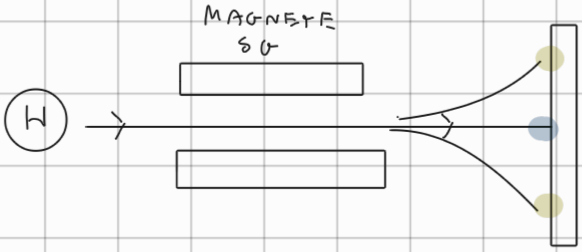
μ_z non quantizzato



2 QUANT.

L_z QUANTIZ.

$l = 2$



$m = 0 \Leftrightarrow l = 0 \Rightarrow$ IPOTESI

REALTÀ 2 macchie distinte

1. Magari $l \neq 0$ Ma dovrei avere num. di macchie dispari X
2. Magari dovuto alle cariche nel nucleo

$$|\vec{\mu}_B| \propto \mu_B \Rightarrow \mu_B^N = \frac{e\hbar}{2m_p} \rightarrow 10^{-3} \text{ volte più piccolo di } \mu_B^e$$ X

$$\Rightarrow F_z = \mu_z \frac{\partial B_z}{\partial z}$$

⌈MOMENTO di DIP. MAGN. di SPIN⌋ → Mom. ang. intrinseco: SPIN

1) S-G non si può fare solo con e^- → $\vec{F}_L = q\vec{v} \wedge \vec{B}$
2) S-G è stato fatto con atomi di Ag → c'è solo 1 el. 5s → $l = 0$

⌈1925⌋ → Cos'è lo SPIN?

$\Rightarrow v \gg c$ X SPIN è un grado di lib. quantistico che non ha analogo classico

$$\vec{F}_z = \mu_z \boxed{\frac{\partial B_z}{\partial z}} \rightarrow \text{Noto (progetto io il magnete)}$$

Misuro la deflessione

RISULTATO $\Rightarrow$ $\boxed{\mu_z = \pm \mu_B}$ $\rightarrow$ Mom. di dip. mag. intrinseco quantizzato per e^-

- $\vec{\mu}_\ell = -g_\ell \, \mu_B \dfrac{\vec{L}}{\hbar} \Leftrightarrow \vec{\mu}_s = -g_s \, \mu_B \dfrac{S}{\hbar}$ $\rightarrow$ (INTRODUCO per analogia)

- $(\mu_\ell)_z = -g_\ell \, \mu_B \, m_\ell \Leftrightarrow (\mu_s)_z = -g_s \, \mu_B \, m_s$

$\left[\hat{\vec{S}} \Leftrightarrow \hat{\vec{L}} \right]$

$\hat{\vec{S}}$ $\rightarrow$ Op. di SPIN

(Si può dim. che:) QUALUNQUE MOM. ANG. $\vec{J}$:

$\Rightarrow [\hat{J}^2, J_i] = 0$ e $[\hat{J}_i, \hat{J}_j] = \varepsilon_{ijk} \hbar \hat{J}_k$

(corso di SIMMETRIE)

$J^2 = \hbar^2 j(j+1) \rightarrow j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2} \ldots$

$m_j = -j, -j+1, \ldots +j$

($\hat{L}$ è un sottoins. dei mom. J generali)

- $[\hat{L}^2, \hat{L}_x] = [\hat{L}^2, \hat{L}_y] = [\hat{L}^2, \hat{L}_z] = 0 \Leftrightarrow [\hat{S}^2, \hat{S}_x] = \ldots = 0$

- $[\hat{L}_i, \hat{L}_j] = \varepsilon_{ijk} i\hbar \hat{L}_k \Leftrightarrow [\hat{S}_i, \hat{S}_j] = \varepsilon_{ijk} i\hbar \hat{S}_k$

- $\hat{L}^2 = \hbar^2 \ell(\ell+1) \Leftrightarrow S^2 = \hbar^2 s(s+1)$
- $L_z = \hbar m_\ell \Leftrightarrow S_z = \hbar m_s$
- $m_\ell = -\ell, \ldots, +\ell \Leftrightarrow m_s = -s, \ldots, +s$

$\Rightarrow (\mu_s)_z = \underbrace{\pm \mu_B}_{\text{Sperimentale}} = -g_s \, \mu_B \, m_s$ $\rightarrow$ $m_s = +\frac{1}{2}$, $m_s = -\frac{1}{2}$ } Due macchie $\Rightarrow g_s = 2$

Se $m_s = 1 \Rightarrow m_s = 0, -1 \Rightarrow$ n. di macchie dispari (m_s salta di 1)

$m_s = -\frac{1}{2}, \frac{1}{2} \Rightarrow s = \frac{1}{2} \Rightarrow$ Num. quant. di spin è semintero

La teor. di SCHRODINGER non PREVEDE lo SPIN

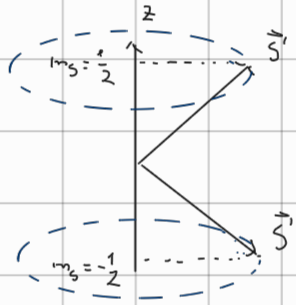
[1918] DIRAC Teoria RELATIVISTICA ; EQ. di DIRAC $\Rightarrow$ Sch. + $E = \sqrt{p^2c^2 + m^2c^4} + V$

$\rightarrow$ Si deriva naturalm. dall'eq. lo spin

SPIN $\Leftrightarrow$ RELATIVITÁ

- Elett., Prot., Neut. $\Rightarrow s = \frac{1}{2}$
- MESONI π (quark + antiquark) $= S = 0$
- Fotoni $\Rightarrow S = 1$
- BARIONI Δ (3 quark) $s = \frac{3}{2}$

ELETTRONI: $S = \frac{1}{2}$, $m_s = \pm\frac{1}{2} \Rightarrow S_z = \pm\frac{\hbar}{2}$



$\Rightarrow$ SPAZIO di HILBERT 2D : $\vec{w}_1\begin{pmatrix}1\\0\end{pmatrix}$, $\vec{w}_2\begin{pmatrix}0\\1\end{pmatrix}$

$\vec{w} = \alpha\,\vec{w}_1 + \beta\,\vec{w}_2$, $|\alpha|^2 + |\beta|^2 = 1$

$|w\rangle = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$ $\quad \langle w| = \begin{pmatrix} w_1^* & w_2^* \end{pmatrix}$ ORA GLI OP. SONO MATRICI

$\hat{Q} \Rightarrow \hat{Q} = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{21} \end{bmatrix}$

$\hat{Q}^+ \Rightarrow \langle \hat{Q}^+ v | w \rangle = \langle v | \hat{Q} w \rangle$

$\Rightarrow \hat{S}^2, \hat{S}_x, \hat{S}_y, \hat{S}_z \Rightarrow$ • $[\hat{S}^2, \hat{S}_x] = [\hat{S}^2, \hat{S}_y] = [\hat{S}^2, \hat{S}_z] = 0$

• $[\hat{S}_i, \hat{S}_j] = i\hbar\,\varepsilon_{ijk}\,\hat{S}_z$

SPINORI: Stato del sistema = FUNZIONE D'ONDA

- $\hat{S}_z \Rightarrow [\hat{S}^2, \hat{S}_z] = 0$

AUTOVETTORI di: $\hat{S}_z$

$\longrightarrow$ SPIN-UP $\Rightarrow \chi_+ = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow S_z = \hbar/2$

$\longrightarrow$ SPIN-DOWN $\Rightarrow \chi_- = |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

SPINORE GENERICO: $|\chi\rangle = a|\uparrow\rangle + b|\downarrow\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$, $|a|^2 + |b|^2 = 1$

$\Rightarrow \langle\chi| = (a^* \; b^*)$

$\Psi = \sum_n \underbrace{c_n}_{(a,b)} \varphi_n$, $|a|^2 = \mathcal{P}\left(S_z = \frac{\hbar}{2}\right)$, $|b|^2 = \mathcal{P}\left(S_z = -\frac{\hbar}{2}\right)$

$\Rightarrow \hat{S}_z = \hbar m_s = \pm\frac{\hbar}{2}$ $\quad \hat{S}_z|\uparrow\rangle = S_z\begin{pmatrix} 1 \\ 0 \end{pmatrix} = +\frac{\hbar}{2}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = +\frac{\hbar}{2}|\uparrow\rangle$

$\hat{S}_z|\downarrow\rangle = \hat{S}_z\begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{\hbar}{2}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{\hbar}{2}|\downarrow\rangle$

$\Rightarrow \boxed{\hat{S}_z = \frac{\hbar}{2}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}}$ $\Rightarrow$ MATRICE DIAGONALE

$s = \frac{1}{2} \Rightarrow S^2 = \hbar^2 s(s+1) = \frac{3}{4}\hbar^2$

$$\begin{cases} \hat{S}^2|\uparrow\rangle = \hat{S}^2\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{3}{4}\hbar^2\begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \hat{S}^2|\downarrow\rangle = \hat{S}^2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{3}{4}\hbar^2\begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{cases} \Rightarrow \boxed{\hat{S}^2 = \frac{3}{4}\hbar^2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}$$

$\Rightarrow$ $\hat{S}_x = \frac{\hbar}{2}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\hat{S}_y = \frac{\hbar}{2}\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ (a diff. di z MAT. NON DIAGONALI)

MATRICI di PAULI:

$$\hat\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \hat S_x = \frac{\hbar}{2}\hat\sigma_x \quad , \quad \hat\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \Rightarrow \hat S_y = \frac{\hbar}{2}\hat\sigma_y$$

$$\hat\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow \hat S_z = \frac{\hbar}{2}\hat\sigma_z$$

(Queste matrici sono legati al modo di ruotare dei vett. nell'alg. di Lie)

$\Rightarrow$ MAT. HERMITIANE e UNITARIE

$\hookrightarrow \left(\hat\sigma_j^+ \hat\sigma_j = 1 \right)$

$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (AUTOST. di $\hat S_z$) è autost. di $\hat S_x$?

$$\hat S_x = \frac{\hbar}{2}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \overset{\hat S_x}{\frac{\hbar}{2}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}} \overset{|\uparrow\rangle}{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} = \frac{\hbar}{2}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{\hbar}{2}|\downarrow\rangle$$

$\Rightarrow$ NON È AUTOSTATO di $\hat S_x$ (non ti dà se stesso)

Trovo autost. di $\hat S_x$:

$$\det\left(\hat S_x - \lambda \mathbb{I}\right) = \begin{vmatrix} -\lambda & \hbar/2 \\ \hbar/2 & -\lambda \end{vmatrix} = 0 \qquad \lambda^2 - \left(\frac{\hbar^2}{2}\right) = 0 \Rightarrow \boxed{\lambda = \pm\frac{\hbar}{2}} \text{ Autov. di } \hat S_x$$

$\rightarrow$ AUTOVET:
$$\begin{bmatrix} -\lambda & \hbar/2 \\ \hbar/2 & -\lambda \end{bmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 0 \qquad \lambda = +\hbar/2$$

$$\rightarrow \begin{bmatrix} -\hbar/2 & \hbar/2 \\ \hbar/2 & -\hbar/2 \end{bmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 0 \qquad -\frac{\hbar}{2}a + \frac{\hbar}{2}b = 0 \rightarrow a = b \Rightarrow |\chi_1\rangle = \begin{pmatrix} 1/\sqrt2 \\ 1/\sqrt2 \end{pmatrix}$$

UN AUTOV. di $\hat S_x$

$$\lambda = -\hbar/2 \rightarrow \begin{bmatrix} \hbar/2 & \hbar/2 \\ \hbar/2 & \hbar/2 \end{bmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 0 \qquad \hbar/2\, a + \frac{\hbar}{2} b = 0$$
$$a = -b \qquad \Rightarrow |\chi_2\rangle = \begin{pmatrix} 1/\sqrt2 \\ -1/\sqrt2 \end{pmatrix}$$

$$\Rightarrow [\hat S_z, \hat S_x] \neq 0$$

MISURA delle COMPONENTI dello SPIN:

$|\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$

$$\langle \hat{S}_z \rangle = \langle \chi | \hat{S}_z | \chi \rangle = \underbrace{(a^* \ b^*)}_{\langle \chi |} \underbrace{\frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}}_{\hat{S}_z} \underbrace{\begin{pmatrix} a \\ b \end{pmatrix}}_{|\chi\rangle} = \frac{\hbar}{2} \left(|a|^2 - |b|^2 \right)$$

$$\Downarrow$$

$$\left(\int \psi^* \hat{S}_x \psi \right)$$

$$\langle \hat{S}_z \rangle = |a|^2 \frac{\hbar}{2} + |b|^2 \left(-\frac{\hbar}{2} \right)$$

$\left(\langle A \rangle = \sum_n P A_n \right)$ Solito def. di media di una grandezza

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$|\chi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\quad \langle \hat{S}_z \rangle = (1 \ 0) \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2}$ (come al solito 100%)

$\rightarrow \langle \hat{S}_x \rangle = \langle \chi | \hat{S}_x | \chi \rangle = (1 \ 0) \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0 \Rightarrow \langle \hat{S}_x \rangle = 0$

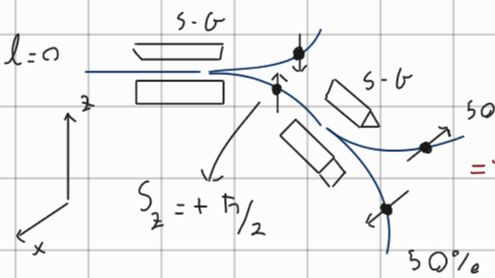
Posso scrivere $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ come sovrap. di autos. di $\hat{S}_x \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} = a \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} + b \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$

$\Rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \Rightarrow$ Se sono in uno stato $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ 50% misuro $|\uparrow\rangle$ 50% $|\downarrow\rangle \Rightarrow \langle \hat{S}_x \rangle = 0$

(Se sono in un autos. di $\hat{S}_z$ non posso conosc. con prec. S_x e S_y)

$(\mu_s)_z = -g_s \mu_B m_s$ $\quad m_s = -\frac{1}{2} \Rightarrow (\mu_s)_z > 0$

$m_s = +\frac{1}{2} \Rightarrow (\mu_s)_z < 0$



$\Rightarrow \langle \hat{S}_x \rangle = 0$

$F \propto (\mu_s)_z$

Il mom. di dip. mag. è inver. risp. allo spin

Se mettessi di nuovo un SG dopo la mis. di $\hat{S}_x$ troverei $\hat{S}_z$ al 50%

$$\hat{\vec{S}} = \frac{\hbar}{2}\hat{\vec{\sigma}} = \frac{\hbar}{2}(\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$$

ATOMO ID.:

$$u_{nlm}(\vec{r}) \rightarrow q = \{n, l, m\} \Rightarrow u_q(\vec{r}) \in \mathcal{L}^2(\mathbb{R}^3)$$

$$|\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \quad a, b \in \mathbb{C} \quad , |a|^2 + |b|^2 = 1$$ MISTO INS.

$$\Rightarrow u(\vec{r}, S_z) = \begin{pmatrix} a\, u_{q_1}(\vec{r}) \\ b\, u_{q_2}(\vec{r}) \end{pmatrix} \Rightarrow \int |a\, u_{q_1}(\vec{r})|^2 d\vec{r} + \int |a\, u_{q_2}(\vec{r})|^2 d\vec{r} = 1$$

Es.:

a. $u(\vec{r}, S_z) = u_{100}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow 1s \boxed{\uparrow}$

b. $u(\vec{r}, S_z) = u_{100}\left[\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right] \Rightarrow 1s \underset{50\%}{\boxed{\uparrow}} \underset{50\%}{\boxed{\downarrow}}$

c. $u(\vec{r}, S_z) = \frac{1}{\sqrt{2}} u_{100}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} u_{200}\begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow 1s \underset{50\%}{\boxed{\uparrow}}, \; 2s \underset{50\%}{\boxed{\downarrow}}$

SOMME di MOMENTI ANGOLARI:

$\vec{L} \Rightarrow L^2 = \hbar^2 l(l+1) \qquad L_z = \hbar\, m_l \qquad m_l = -l \dots l$

$\hat{S} \Rightarrow S^2 = \hbar^2 s(s+1) \qquad S_z = \hbar\, m_s \qquad m_s = \pm\frac{1}{2}$

GENERICO $\vec{J} \Rightarrow [\hat{J}^2, \hat{J}_x] = [\hat{J}^2, \hat{J}_y] = [\hat{J}^2, \hat{J}_z] = 0 \;, [\hat{J}_i, \hat{J}_j] = i\hbar\varepsilon_{ijk}\hat{J}_k$

$J^2 = \hbar^2 j(j+1) \qquad J_z = \hbar\, m_j \qquad m_j = -j \dots j$

$$\underset{\substack{\downarrow \\ (j_1, m_{j_1})}}{\vec{J}_1} + \underset{\substack{\downarrow \\ (j_2, m_{j_2})}}{\vec{J}_2} = \vec{J}$$

(per fare la somma dovrei conoscere tutte le componenti)

TEOREMA di SOMMA dei MOMENTI ANGOLARI:

j (num. quant.) $= |j_1 - j_2| \ldots\ldots j_1 + j_2$ SALTANDO di 1

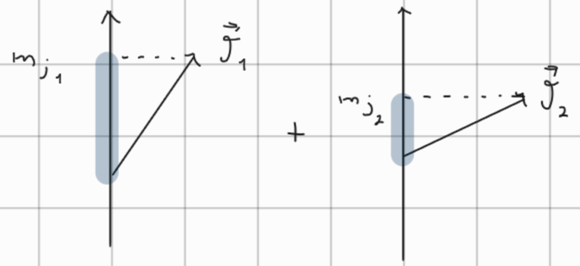
Es:

$$j_1 = 1 \qquad j_2 = 3 \quad \Rightarrow \quad j = 2, 3, 4$$

$j = 2 \rightarrow m_j = -2, \ldots, 2$

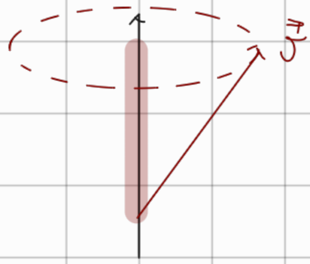
$j = 3 \rightarrow m_j = -3, \ldots, 3$

$j = 4 \rightarrow m_j = -4, \ldots, 4$



Posso sommare le proiez.

$\Rightarrow$



$m_j = m_{j_1} + m_{j_2}$

$\max m_{j_1} = j_1$

$\max m_{j_2} = j_2$

$\max m_j = j_1 + j_2 = j$

$\vec{J}_1$: $[\hat{J}_1^2, J_{1i}] = 0 \qquad i = x, y, z$

$[\hat{J}_{1i}, \hat{J}_{1j}] = i\hbar\, \varepsilon_{ijk}\, \hat{J}_{1k}$

$\vec{J}_2$: stesso cosa

$\vec{J}_1$ e $\vec{J}_2$ COMPLETAM. INDIPEND.

$[\hat{J}_1^2, \hat{J}_2^2] = 0$

$[\hat{J}_1^2, \hat{J}_{2i}^2] = [\hat{J}_2^2, \hat{J}_{1i}^2] = 0$

$[\hat{J}_{1i}, \hat{J}_{2k}] = 0 \qquad i, k = x, y, z$

$\Rightarrow \hat{J}_1^2, \hat{J}_{1z}, \hat{J}_2^2, \hat{J}_{2z}$ COMMUTANO MUTUAMENTE (HANNO UNA BASE di AUTOVET. in COMUNE)

BASE DISACCOPPIATA

$\Rightarrow \boxed{|j_1, m_{j_1}, j_2, m_{j_2}\rangle}$ (indico l'autostato tramite i num. quant.)

OP. AUTOF. AUTOV. AUTOF.

$$\Rightarrow \hat{J}_1^2 |j_1, m_{j_1}, j_2, m_{j_2}\rangle = \hbar^2 j_1(j_1+1) |j_1, m_{j_1}, j_2, m_{j_2}\rangle$$

$$\hat{J}_{1z} |j_1, m_{j_1}, j_2, m_{j_2}\rangle = \hbar m_{j_1} |j_1, m_{j_1}, j_2, m_{j_2}\rangle$$

$$\hat{J}_2^2 |j_1, m_{j_1}, j_2, m_{j_2}\rangle = \hbar^2 j_2(j_2+1) |j_1, m_{j_1}, j_2, m_{j_2}\rangle$$

$$\hat{J}_{2z} |j_1, m_{j_1}, j_2, m_{j_2}\rangle = \hbar m_{j_2} |j_1, m_{j_1}, j_2, m_{j_2}\rangle$$

$\Rightarrow \hat{H} w_{nlm} = E_n w_{nlm} \Rightarrow \hat{H}|nlm\rangle = E_n |nlm\rangle$

Es. $\psi = \frac{1}{\sqrt{2}} w_{100} + \frac{1}{\sqrt{2}} w_{210} = \frac{1}{\sqrt{2}}|100\rangle + \frac{1}{\sqrt{2}}|210\rangle$

$L^2 Y_1^0 = \hbar^2 l(l+1) Y_1^0 \Rightarrow L^2|lm\rangle = \hbar^2 l(l+1)|lm\rangle$

$\hat{J}^2 = (\hat{J}_1 + \hat{J}_2)^2 = \hat{J}_1^2 + \hat{J}_2^2 + 2 \underbrace{\hat{J}_1 \hat{J}_2}_{J_{1x}J_{2x} + J_{1y}J_{2y} + J_{1z}J_{2z}}$

$[\hat{J}^2, \hat{J}_1^2] = [\hat{J}^2, \hat{J}_2^2] = 0$

$[\hat{J}_1^2, \hat{J}_z] = [\hat{J}_2^2, \hat{J}_z^2] = 0$ VERIFICO DOPO

$[\hat{J}^2, \hat{J}_{1z}] \neq 0 \quad [\hat{J}^2, \hat{J}_{2z}] \neq 0$

$\hat{J}^2, \hat{J}_z, \hat{J}_1^2, \hat{J}_2^2 \Rightarrow \boxed{|j, m_j, j_1, j_2\rangle}$ BASE ACCOPPIATA

COMMUTANO TRA DI LORO

$\Rightarrow \hat{J}^2|j, m_j, j_1, j_2\rangle = \hbar^2 j(j+1)|j, m_j, j_1, j_2\rangle$

$\hat{J}_z|j, m_j, j_1, j_2\rangle = \hbar m_j|j, m_j, j_1, j_2\rangle$

$\hat{J}_1^2|j, m_j, j_1, j_2\rangle = \hbar^2 j_1(j_1+1)|j, m_j, j_1, j_2\rangle$

$\hat{J}_2^2|j, m_j, j_1, j_2\rangle = \hbar^2 j_2(j_2+1)|j, m_j, j_1, j_2\rangle$

COEF. DI CLEBSCH GORDAN

$|j, m_j, j_1, j_2\rangle = \sum_{m_{1j}, m_{2j}} C_{m_{1j}, m_{2j}} |j_1, m_{1j}, j_2, m_{2j}\rangle$

SE I VETT. SONO IND. POSSO PASSARE DA UNA BASE ALL'ALTRA

SPIN DI DUE ELETTRONI:

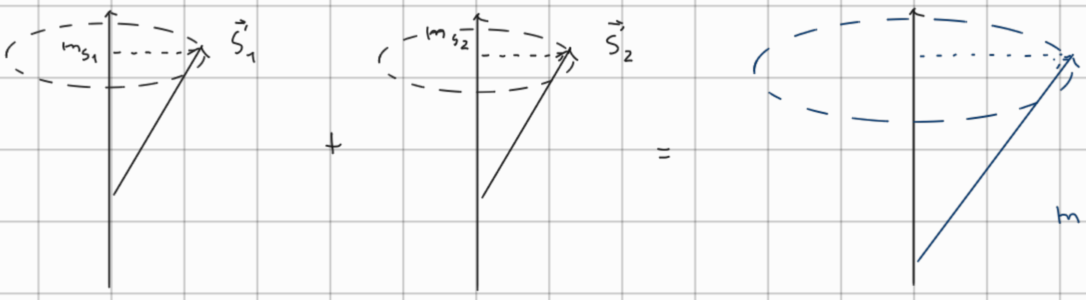
$\begin{cases} S_1 = \frac{1}{2} \\ S_2 = \frac{1}{2} \end{cases}$ $\quad S = |S_1 - S_2|, \dots, S_1 + S_2 \Rightarrow S = 0, 1$

SE $S = 0 \Rightarrow m_s = 0$ SINGOLETTO

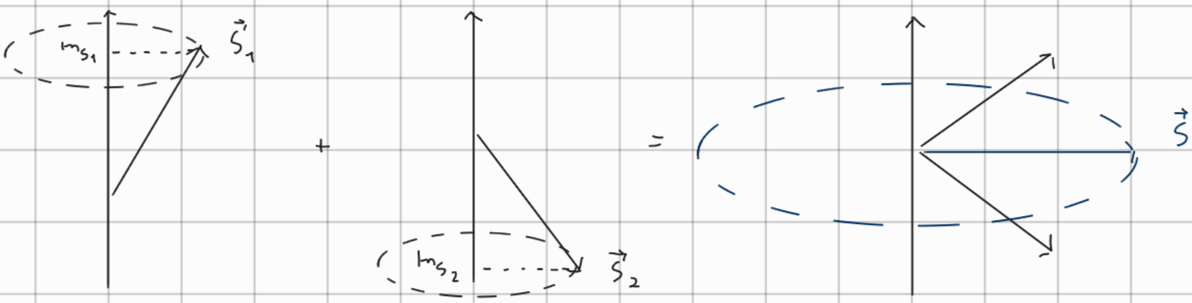
SE $S = 1 \Rightarrow m_s = -1, 0, 1$ TRIPLETTO

$$m_{s_1} = -\frac{1}{2}, +\frac{1}{2}$$
$$m_{s_2} = -\frac{1}{2}, +\frac{1}{2}$$

$$\Rightarrow \quad m_s = m_{s_1} + m_{s_2} = \begin{cases} 1 & \uparrow \uparrow \\ 0 & \uparrow \downarrow \\ 0 & \downarrow \uparrow \\ -1 & \downarrow \downarrow \end{cases}$$



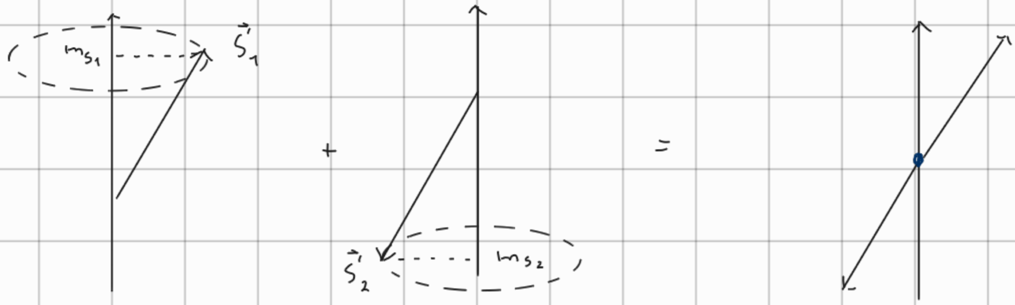
$m_s = 1$ $\boxed{\uparrow\uparrow}$



$m_s = 0$ $\boxed{\uparrow\downarrow} = \boxed{\downarrow\uparrow}$

Uguale per $\downarrow + \downarrow = \boxed{\downarrow\downarrow}$

$\Rightarrow$ SINGOLETTO $S = 0$, $m_s = 0$



$m_s = 0$ $S = 0$

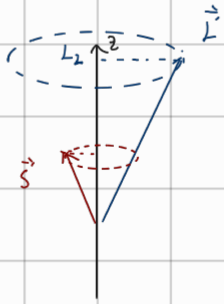
MOMENTO ANG. TOTALE:

- $e \longrightarrow \vec{L}$ $\quad \vec{J} = \vec{L} + \vec{S}$ $\quad j = |l - s|, \dots, l + s = |l - \frac{1}{2}|, \dots, l + \frac{1}{2}$

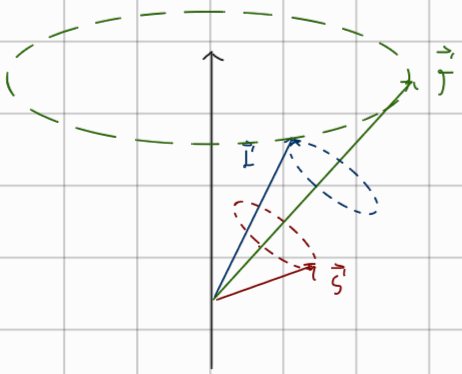
  $\quad \llcorner \; \vec{S}$

  POSSO USARE ENTRAMBI BASI ACCOPP. E DISACC.

  $\rightarrow |l, m_l, s, m_s\rangle$ DISACCOPPIATA



$\to |j, m_j, \ell, s\rangle$ BASE ACCOPPIATA



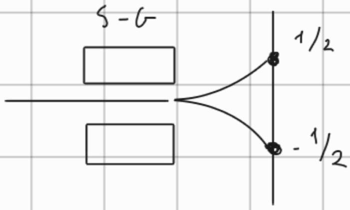
$\Rightarrow$ Da questo dis. si capisce perché L_z e S_z non sono definiti (m_ℓ e m_s non compaiono nella base)

$$\vec{\mu}_j = -g_j \mu_B \frac{\vec{J}}{\hbar} \qquad g_\ell = 1 \;, \; g_s = 2 \qquad g_j = 1 + \frac{j(j+1) - \ell(\ell+1) + s(s+1)}{2j(j+1)}$$

$$(\mu_j)_z = -g_j \mu_B m_j$$

FATTORE di LANDÉ

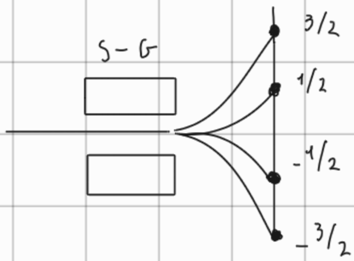
Es.: • H $\quad 1s$ $\boxed{\uparrow}$ $\quad \ell = 0, \; s = \frac{1}{2} \Rightarrow j = \frac{1}{2} \Rightarrow g_j = g_s = 2 \;, \quad m_j = m_s = \pm \frac{1}{2}$



• H $\boxed{\uparrow \,|\, \;|\; }$ $\quad 2p \quad \ell = 1, \; s = \frac{1}{2} \Rightarrow j = \frac{3}{2}, \; \frac{1}{2}$

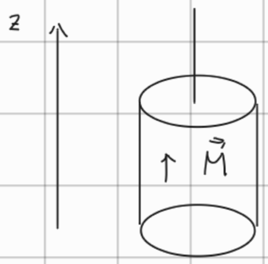
$$j = \frac{3}{2} \Rightarrow m_j = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$$

$$(\mu_j)_z = -g_j \mu_B m_j$$



L'esperimento conferma la teoria

EFFETTO EINSTEIN-DE HASS



dens. di dip. mag.

$\vec{M} = n\vec{\mu}$

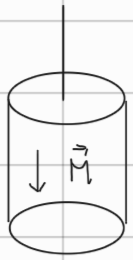
$\langle \vec{L} \rangle = 0$ QUENCHING di $\vec{L}$

$\vec{M} = n\vec{\mu}_s \Rightarrow S_z = -\frac{\hbar}{2}$ (μ_s è opposto al campo mag.)

$(\mu_s)_z = -g_s \mu_B m_s$

$M = -n g_s \mu_B m_s$

$\Rightarrow$



$S_z = -\frac{\hbar}{2} \Rightarrow S_z = \frac{\hbar}{2} \rightarrow \Delta M = n g_s \mu_B \frac{\Delta S_z}{\hbar} = n g_s \mu_B \underbrace{\Delta m_s}_{1}$

IL MOM. ANG. TOT. SI DEVE CONS. (SIST. ISOLATO)

$\Delta J_z = \Delta L_z + \Delta S_z = 0$

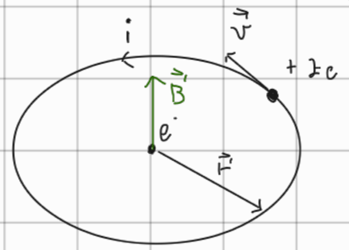
$\Downarrow L_z = -\Delta S_z = -\frac{\hbar \Delta M}{n g_s \mu_B} \hat{n}_z$

SE CAMBIO LA MAGNETIZ. IL CILINDRO RUOTA

$w = \frac{L_z}{I}$

INTERAZIONE di SPIN-ORBITA:

sono nel sis. di rif. di e^-

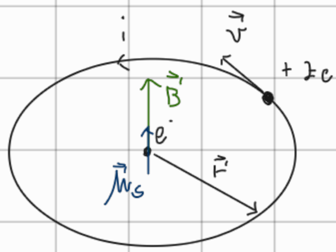


$\vec{B} = \frac{\mu_0 i}{2r}\hat{u}_\perp$, $i = \frac{Ze}{T} = \frac{Ze}{\frac{2\pi r}{v}} = \frac{Zev}{2\pi r}$

$|\vec{B}| = \frac{\mu_0 Zev}{4\pi r^2} \frac{m_e r}{m_e r} = \frac{\mu_0 Ze}{4\pi m r^3} mvr \cdot \frac{\varepsilon_0}{\varepsilon_0} =$

$= \frac{\overbrace{\varepsilon_0 \mu_0}^{1/c^2} Ze}{4\pi\varepsilon_0 m r^3} mvr = \frac{Ze}{4\pi\varepsilon_0 c^2 m r^3} mvr$

$|\vec{B}| = \frac{Ze}{4\pi\varepsilon_0 c^2 m r^3} |\vec{L}| \Rightarrow \boxed{\vec{B} = \frac{Ze}{4\pi\varepsilon_0 c^2 m r^3} \vec{L}}$

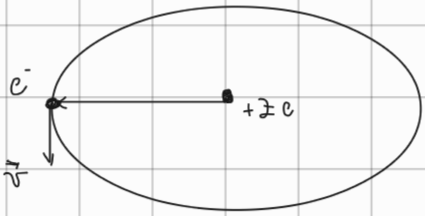


CONSIDERO $\vec{\mu}_s$:

$V = -\vec{\mu}_s \cdot \vec{B}$ (HO TROVATO QUALCOSA CHE ACCOPPIA LO SPIN E LE ORBITE)

$\Downarrow$ dipende da s → Dipende da ℓ

$$\Rightarrow \boxed{\Delta E_{SO} = -\vec{\mu}_s \cdot \vec{B}}$$



Per tornare con il prot. in quiete servono le trasform. di LORENTZ. Se si fanno i conti:

$$\boxed{\Delta E_{SO} = -\frac{1}{2} \vec{\mu}_s \cdot \vec{B}}$$

↳ Precessione di Thomas

$$\Delta E_{SO} = -\frac{1}{2} \vec{\mu}_s \cdot \vec{B} = \frac{1}{2} g_s \mu_B \frac{S}{\hbar} \cdot \frac{Ze}{4\pi \varepsilon_0 c^2 m r^3} \vec{L}$$

$$\boxed{\Delta E_{SO} = \frac{Ze^2}{8\pi \varepsilon_0 m^2 c^2 r^3} \vec{S} \cdot \vec{L}}$$

$$\Rightarrow \quad S \sim L \sim \hbar \quad , \quad r \sim a_0 = \frac{4\pi \varepsilon_0 \hbar^2}{m e^2} \quad , \quad E \sim Ry = \frac{m e^4}{2(4\pi \varepsilon_0 \hbar)^2}$$

$$\Delta E_{SO} \simeq \frac{Ze^2}{8\pi \varepsilon_0 m^2 c^2 \left(\frac{4\pi\varepsilon_0 \hbar^2}{m e^2}\right)^3} \cdot \hbar^2 = \frac{Z m^3 e^8 \hbar^2}{8\pi (4\pi)^3 \varepsilon_0^4 m^2 c^3 \hbar^6} = \frac{Z m e^8}{2(4\pi)^4 \varepsilon_0^4 c^2 \hbar^4} =$$

$$= Z \underbrace{\left[\frac{m e^4}{2(4\pi\varepsilon_0 \hbar)^2}\right]}_{Ry} \underbrace{\left[\frac{e}{4\pi \varepsilon_0 c \hbar}\right]^2}_{\alpha \equiv \frac{1}{137}} = Z\, Ry\, \alpha^2 \Rightarrow \frac{\Delta E}{E} \simeq \alpha^2 \simeq 10^{-4}$$

Variaz. piccole ma non trasc.

$$\Delta E_{SO} = \frac{1}{2m^2c^2} \underbrace{\frac{Ze^2}{4\pi \varepsilon_0 r^2}}_{F_{el} = -|\nabla V| = \frac{dV}{dr}} \vec{S} \cdot \vec{L} \Rightarrow \boxed{\Delta E_{SO} = \frac{1}{2m^2c^2} \left(\frac{1}{r} \frac{dV}{dr}\right) \vec{S} \cdot \vec{L}}$$

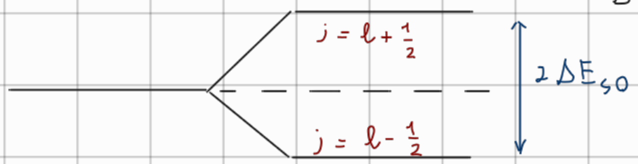
$$\hat{H}_{SO} = \frac{Ze^2}{8\pi \varepsilon_0 m^2 c^2 r^3} \vec{S} \cdot \vec{L} \Rightarrow \boxed{\hat{H} = -\frac{\hbar^2}{2m} \Delta + \hat{V}(r) + \hat{H}_{SO}}$$

(NON C'È UNA SOL. ANALITICA DIRETTA)

Lo risolv. con la teor. delle pert. non dipend. dal tempo

→ Si UTILIZZA la BASE ACCOPPIATA (s e l sono legati) $\Rightarrow |j, m_j, l, s\rangle$

$\vec{L}\uparrow \Rightarrow \vec{B}\uparrow$; $\vec{\mu}_s \downarrow$ $\quad \Delta E_{so} = -\vec{\mu}_s \cdot \vec{B} \Rightarrow$ POSITIVO (HO UN AUM. DELL'EN.)



DOPPIETTI di SPIN ORBITA

$\vec{L}\uparrow = \vec{B}\uparrow$; $\vec{\mu}_s \uparrow$ ($s\downarrow$) $\quad \Delta E_{so}$ é NEGATIVA (stato di En. minore)

REGOLE di SELEZIONE per $\vec{J}$:

Elet. : $s = j = 1$ (l'elet. ha un mom. ang. intrinseco e ang.)

- POSTULATO: LA LUCE viaggia a c in tutti i sistemi di riferimento

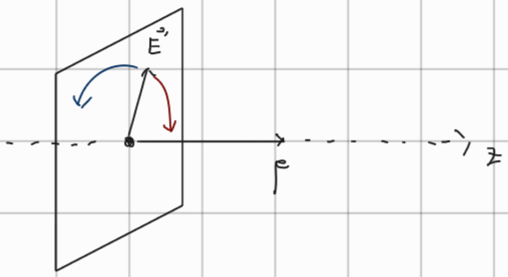
$\Rightarrow$ Non posso dist. mom ang. e intrinseco

$\Rightarrow J^2_{fot} = S^2_{fot} = \hbar^2 s(s+1) \Rightarrow s = 1 \rightarrow m_s = +1, 0, -1$

① Se $M = 0$ (massa) $\Rightarrow$ l'unica direz. di s accettabile é quella // a $\vec{p}$

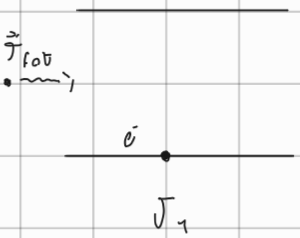
$\cdots\cdots \bullet\!\longrightarrow \vec{p} \cdots\cdots\rightarrow z \Rightarrow$ Asse di quant.

② CAMPI E.M. $\Rightarrow$ Trasversale $\Rightarrow$ $\cancel{m_s = 0}$



$m_s = +1 \quad S_z = J_z = \hbar \Rightarrow$ Polariz. circ. destra

$m_s = -1 \quad S_z = J_z = -\hbar \Rightarrow$ pol. circ. sinistra



$\vec{J}_1 + \vec{J}_{fot} = \vec{J}_2 \quad (j_{fot} = 1)$

$\Rightarrow j_2 = |j_1 - 1|, j_1, j_1 + 1$

$\Rightarrow \boxed{\Delta j = 0, \pm 1}$ REGOLA di SELEZ. per $\vec{J}$

(in processo di ass. o emis. di un fot)

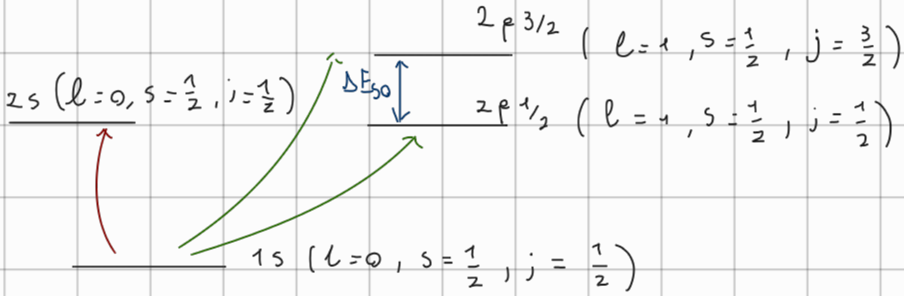
$\boxed{S=\frac{1}{2}} \Rightarrow \boxed{\Delta S = 0}$ in una trans. di dip. elet.

$\rightarrow$ Quindi il ± 1 del fot. finisce su ℓ, ecco pk

$\ulcorner \Delta \ell = \pm 1 \lrcorner$

REGOLE di SELEZ. di DIPOLO ELETTRICO:
$$\begin{cases} \Delta \ell = \pm 1 \\ \Delta j = 0, \pm 1 \\ \Delta S = 0 \end{cases}$$

Es.



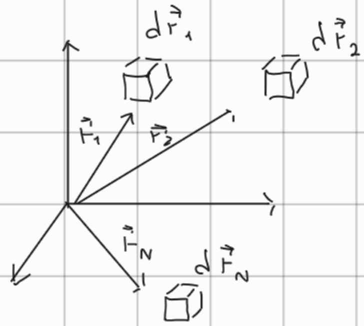
$1s \to 2s : \Delta j = 0, \Delta \ell = 0$ ✗ $\qquad 1s \to 2p_{3/2} : \Delta j = 1, \Delta \ell = 1$ ✓

$1s \to 2p_{1/2} : \Delta j = 0, \Delta \ell = +1$ ✓

SISTEMI a MOLTE PARTICELLE:

$\Rightarrow$ Per ora TRASCURIAMO lo SPIN

N particelle $\quad \psi_{TOT} = \psi(\vec{r}_1, \vec{r}_2, \vec{r}_3 \dots \vec{r}_N, t)$



$|\psi_{TOT}|^2 \to P$ CONGIUNTA All' ist. t

$\rightarrow$ Part. 1 in $d\vec{r}_1$

$\rightarrow$ Part. 2 in $d\vec{r}_2$

$\Rightarrow i\hbar \frac{\partial \psi_{TOT}}{\partial t} = \hat{H}\psi_{TOT} \quad , \quad \hat{H} = \hat{K}_{TOT} + \hat{V}(\vec{r}_1, \vec{r}_2, \vec{r}_3, \dots t)$

$\rightarrow$ Se $\hat{V} \neq \hat{V}(t) \qquad \psi_{TOT} = \underline{u(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)}\, e^{-i\frac{E_{TOT}}{\hbar}t}$

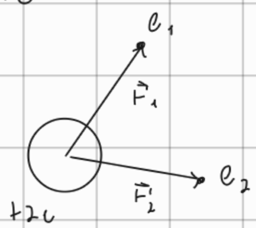
$\hookrightarrow$ Non esiste una soluz. esatta (a partire da $N=3$)

$\Rightarrow \ulcorner \hat{H}_{TOT}\, u = E_{TOT}\, u \lrcorner$

⇒ APPROSSIMAZIONE di PARTICELLE INDIPENDENTI:

→ Trascuro mutua interazione tra particelle

Es.: He



TERMINE di repulsione

$$\hat H = \overbrace{-\frac{\hbar^2}{2m}\Delta_1 - \frac{\hbar^2}{2m}\Delta_2}^{\hat K_{tot}} \overbrace{-\frac{2e^2}{4\pi\varepsilon_0 r_1}}^{\hat V_1(\vec r_1)} \overbrace{-\frac{2e^2}{4\pi\varepsilon_0 r_2}}^{\hat V_2(\vec r_2)} + \overbrace{\frac{e^2}{4\pi\varepsilon_0|\vec r_1 - \vec r_2|}}^{\hat V_{ee}}$$

(Non risolv. anal.)

⇒ App. part. ind ⇒ $\hat V_{ee} = 0$

$$\Rightarrow \hat H = \sum_i \hat K_i + \sum_i \hat V_i(\vec r_i) \quad \rightarrow \quad \hat H_i = -\frac{\hbar^2}{2m}\Delta_i + V_i(\vec r_i)$$

$$\Rightarrow \hat H\, w(\vec r_1, \vec r_2 \dots \vec r_N) = \sum_i \hat H_i\, w(\vec r_1, \vec r_2, \dots, \vec r_N) = E_{TOT}\, w$$

$$\Rightarrow w(\vec r_1, \vec r_2, \dots, \vec r_N) = w_1(\vec r_1)\, w_2(\vec r_2) \dots w_N(r_N)$$ (stesso rag. della buca di pot. inf.)

$$\Rightarrow \sum_i \hat H_i w(\vec r_1, \vec r_2 \dots \vec r_N) = \sum_i \hat H_i\, w_1(r_1) \dots w_N(\vec r_N) = E_{TOT}\, w_1 w_2 \dots w_N$$

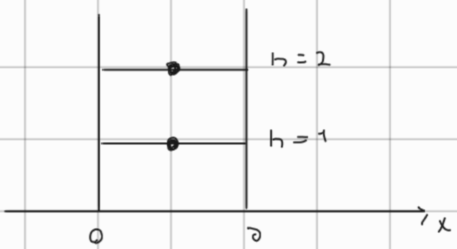
$$\Rightarrow \sum_i \hat H_i\, w_i(\vec r_i) \prod_{j\neq i} w_j(\vec r_j) = E_{TOT} \prod_j w_j$$

⇒ Divido per $w(r_1, r_2, r_N) = \prod_j w_j$

$$\Rightarrow \sum_i \left[\frac{\hat H_i\, w_i(\vec r_i)}{w_i(\vec r_i)}\right] = \overset{COST}{E_{TOT}} \rightarrow \boxed{\hat H_i\, w_i(\vec r_i) = E_i\, w_i(\vec r_i)} \Rightarrow \boxed{E_{TOT} = \sum_i E_i}$$

Sono tornato ad un prob. ad una part.

Es.



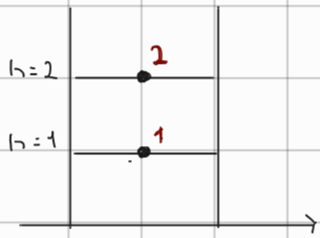
$$w(x_1, x_2) = w_1(x_1)\, w_2(x_2)$$

INDISTINGUIBILITÀ delle PARTICELLE:

$$\psi(\vec r_1, \vec r_2, \dots \vec r_N) = \psi_{\underset{\sim}{1}}(\vec r_1)\ \psi_{\underset{\sim}{2}}(\vec r_2)\ \psi_{\underset{\sim}{N}}(\vec r_N)$$

→ SET di NUM. QUANTICI per descrivere lo STATO $N = \{n, l, m\}$

$\Rightarrow$ Cons. 2 part. $\Rightarrow$ $\psi_1(\vec r_1)\ \psi_2(\vec r_2) = \psi_a(1)\ \psi_b(2)$



$$\psi_a(1)\ \psi_b(2) = \psi_1(x_1)\ \psi_2(x_2)$$

Posso SEMPRE STABILIRE QUALE È 1 e QUALE 2?

1 e 2 non hanno una vera e propria traiettoria, le funzioni d'onda si sovrappongono

$\Rightarrow$ PRINCIPIO di INDISTINGUIBILITÀ delle PARTICELLE

particelle element. della stessa natura sono indistinguibili

$$\psi_a(1)\ \psi_b(2) \xLeftrightarrow[\text{LE PART.}]{\text{SCAMBIO}} \psi_a(2)\ \psi_b(1) \quad , \quad \psi_1(x_1)\ \psi_2(x_2) \Longleftrightarrow \psi_1(x_2)\ \psi_2(x_1)$$

DEVONO entrare in modo paritetico nello stato

$\hat P_{12}$ → OPERATORE di SCAMBIO $\Rightarrow$ $\hat P_{12}\ \psi(1,2) = \psi(2,1)$ Fatt. di fase

→ Fisicamente indistinguibili: $|\psi(2,1)|^2 = |\psi(1,2)|^2 = |e^{i\alpha}\ \psi(1,2)|^2$

→ $\hat P_{12}\ \psi(1,2) = \psi(2,1) = e^{i\alpha}\ \psi(1,2)$

→ $\hat P_{12}^2\ \psi(1,2) = \hat P_{12}\hat P_{12}\ \psi(1,2) = \hat P_{12}\ \psi(2,1) = \hat P_{12}\ e^{i\alpha}\ \psi(1,2) = e^{i\alpha}\ \hat P_{12}\ \psi(1,2) =$

$= e^{i2\alpha}\ \psi(1,2)$

$$\Rightarrow \psi(1,2) = e^{i2\alpha}\ \psi(1,2) \quad \rightarrow \quad \alpha = 0\ ,\ \alpha = \pi$$

$$\boxed{\hat P_{12}\ \psi(1,2) = \pm\ \psi(1,2)}$$ Eq. agli AUTOVAL di $\hat P_{12}$ $\boxed{\lambda = \pm 1}$

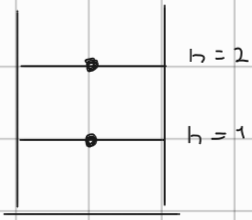
$\Rightarrow \hat{P}_{12}\, \psi(1,2) = \psi(2,1) = \pm\, \psi(1,2)$ Se scambio le part. tra loro

① ψ RESTA UGUALE $\psi(2,1) = \psi(1,2)$

$\Rightarrow \psi_S \to$ FUNZIONE d'onda SIMMETRICA

② ψ CAMBIA SEGNO $\psi(2,1) = -\psi(1,2)$

$\Rightarrow \psi_A \to$ FUNZ. d'ONDA ANTISIMMETRICA



$\psi_1(x_1)\, \psi_2(x_2) \to$ SCAMBIO

$\psi_1(x_2)\, \cancel{\psi_2}(x_1) \Rightarrow$ NON é $\psi(2,1) = \pm\, \psi(1,2)$

$$\Rightarrow \psi_S(1,2) = A\left[\psi_a(1)\, \psi_b(2) + \psi_a(2)\, \psi_b(1)\right]$$

FUNZIONI d'ONDA SIMMETRIZZATE

$$\Rightarrow \psi_A(1,2) = A\left[\psi_a(1)\, \psi_b(2) - \psi_a(2)\, \psi_b(1)\right]$$

l'indistinguibilità produce un obbligo di SIMMETRIZZAZIONE

Per buco di pot: $\psi_S = A\left[\psi_1(x_1)\, \psi_2(x_2) + \psi_1(x_2)\, \psi_2(x_1)\right]$

$$\psi_A = A\left[\psi_1(x_1)\, \psi_2(x_2) - \psi_1(x_2)\, \psi_2(x_1)\right]$$

$$\Rightarrow \hat{P}_{ij}\, \psi(1,2,\dots i,\dots j\dots N) = \psi(1,2,i,j,N) = \pm\, \psi(1,2,i,j,N)$$

QUINDI QUALE SCELGO?

PROPRIETÀ STATISTICHE: $\Rightarrow$ SPIN

TEOR.: SPIN - STATISTICA:

SPIN INTERO $\Rightarrow$ BOSONI (fot., mesoni) $\Rightarrow \psi(1,2) = \psi(2,1)$

SPIN SEMI-INTERO $\Rightarrow$ FERMIONI (elett., neut., prot.) $\Rightarrow \psi(1,2) = -\psi(2,1)$

$$\psi_A(1,2) = \frac{1}{\sqrt{2}}\left[\psi_a(1)\,\psi_b(2) - \psi_a(2)\,\psi_b(1)\right]$$ TRASCURO LO SPIN

$\Rightarrow a = b \Rightarrow \boxed{\psi_A = 0}$ PRINCIPIO di ESCLUSIONE di PAULI

due fermioni identici non possono occupare lo stesso stato quantistico.
(determina la stabilità della materia)

Es:

$$\psi_A = \frac{1}{\sqrt{2}}\left[\psi_1(x_1)\,\psi_2(x_2) - \psi_1(x_2)\,\psi_2(x_1)\right]$$

$1 \to 2 \qquad \psi_A = 0$

nella buca di pot. per ogni stato ci può stare solo un el.

FUNZIONE D'ONDA TOTALE e SCAMBIO:

• 2 FERMIONI IDENTICI:

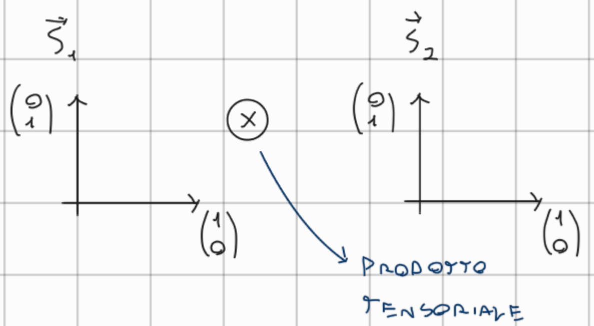
$$\psi(1,2) = -\psi(2,1) \quad , \quad \psi(1,2) = \overset{\text{PARTE SPAZIALE}}{\varphi(1,2)}\; \underset{\text{SPIN}}{\chi(1,2)}$$

$$\rightarrow \varphi(1,2) = \frac{1}{\sqrt{2}}\left[\varphi_a(1)\,\varphi_b(2) \pm \varphi_a(2)\,\varphi_b(1)\right]$$ → APPROS. di PART. IND.

(Basta la presenza degli elettroni per avere correlaz. tra le funz. d'onda)

CORRELAZIONE ⇔ ENTANGLEMENT

$\rightarrow$	S_{1z}	S_{2z}	
$m_s = 1$	$\uparrow \hbar/2$	$\uparrow \hbar/2$	$\rightarrow \chi_+(1)\,\chi_+(2) = \lvert\uparrow\uparrow\rangle$
$m_s = 0$	$\downarrow -\hbar/2$	$\uparrow \hbar/2$	$\rightarrow \chi_-(1)\,\chi_+(2) = \lvert\downarrow\uparrow\rangle$
$m_s = 0$	$\uparrow \hbar/2$	$\downarrow -\hbar/2$	$\rightarrow \chi_+(1)\,\chi_-(2) = \lvert\uparrow\downarrow\rangle$
$m_s = -1$	$\downarrow -\hbar/2$	$\downarrow -\hbar/2$	$\rightarrow \chi_-(1)\,\chi_-(2) = \lvert\downarrow\downarrow\rangle$



$$\Rightarrow \underset{\vec{S}_1}{2D} \otimes \underset{\vec{S}_2}{2D} = 4D$$

$$\vec{v}_1 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \quad \vec{w}_2 = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} , \quad \vec{v}_1 \otimes \vec{w}_2 = \begin{pmatrix} v_1 w_1 \\ v_1 w_2 \\ v_2 w_1 \\ v_2 w_2 \end{pmatrix}$$

$$\Rightarrow |\uparrow\uparrow\rangle \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} , \quad |\downarrow\uparrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$|\uparrow\downarrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} , \quad |\downarrow\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$\Rightarrow$ Autostati della base disaccoppiata $|s_1, m_{1s}, s_2, m_{2s}\rangle$ (non rispettano la simm. per scambio)

$$\hat{S}_{1z} |\uparrow\downarrow\rangle = +\frac{\hbar}{2} |\uparrow\downarrow\rangle , \quad \hat{S}_{2z} |\uparrow\downarrow\rangle = -\frac{\hbar}{2} |\uparrow\downarrow\rangle$$

BASE ACCOPPIATA $\quad \hat{S}^2, \hat{S}_z, \hat{S}_1^2, \hat{S}_2^2$

$$|s, m_s, s_1, s_2\rangle \Rightarrow \hat{S}^2 |s, m_s, s_1, s_2\rangle = \hbar^2 s(s+1) |s, m_s, s_1, s_2\rangle$$

$$\hat{S}_z |s, m_s, s_1, s_2\rangle = \hbar m_s |s, m_s, s_1, s_2\rangle$$

$$\left.\begin{aligned} s_1 &= \tfrac{1}{2} \\ s_2 &= \tfrac{1}{2} \end{aligned}\right\} \quad s = 0, 1$$

$1 \rightarrow$ TRIPLETTO $\quad m_s = -1, 0, 1$

$0 \rightarrow$ SINGOLETTO $\quad m_s = 0$

- AUTOSTATI BASE ACCOPPIATA:

$\boxed{s=0} \rightarrow [m_s = 0] \rightarrow$ SINGOLETTO

$$\chi = \frac{1}{\sqrt{2}} \left[\chi_+(1)\chi_-(2) - \chi_-(1)\chi_+(2) \right]$$

$$\rightarrow |\underset{s}{0}, \underset{m_s}{0}\rangle = \frac{1}{\sqrt{2}} \left[|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \right]$$

Gli stati di SINGOLETTO risultano naturalm. ANTISIMMETRICI PER SCAMBIO

$\boxed{s=1} \rightarrow [m_s = -1, 0, +1] \rightarrow$ TRIPLETTO

$\rightarrow m_s = 1 \quad \chi = \chi_+(1)\chi_+(2) , \quad |1, 1\rangle = |\uparrow\uparrow\rangle$

$\rightarrow m_s = 0 \quad \chi = \frac{1}{\sqrt{2}} \left[\chi_+(1)\chi_-(2) + \chi_-(1)\chi_+(2) \right] , \quad |1, 0\rangle = \frac{1}{\sqrt{2}} \left[|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle \right]$

$\rightarrow m_s = -1 \quad \chi = \chi_-(1)\chi_-(2) , \quad |1, -1\rangle = |\downarrow\downarrow\rangle$

GLI STATI DI TRIPLETTO SONO SIMMETRICI RISPETTO ALLO SCAMBIO

(lavorare con la base accoppiata identifica autom. gli stati simm e antisim. per scambio)

$\vec{S}_1$ e $\vec{S}_2$ INDIPENDENTI $\Rightarrow$ LA BASE DISACCOPPIATA NON TIENE CONTO DELL'INDISTING. delle PARTICELLE

$\Rightarrow$ DEVO USARE LA BASE <u>ACCOPPIATA</u>, devo considerare la simmetrizzaz. per scambio

---

$$\psi(1,2) = \begin{cases} \varphi_A(1,2)\,\chi_{sim}(1,2) \\ \varphi_S(1,2)\,\chi_{ANTI}(1,2) \end{cases} \Rightarrow$$ così la funz. d'onda è globalmente ANTISIMMETRICA

↓ SPIN - STATISTICA

↳ $\psi(1,2) = -\psi(2,1)$

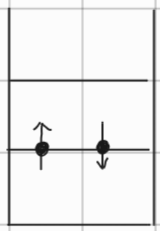
- $\psi(1,2) = \underbrace{\frac{1}{\sqrt{2}}\left[\varphi_a(1)\,\varphi_b(2) + \varphi_a(2)\,\varphi_b(1)\right]}_{\varphi_S(1,2)} \cdot \underbrace{\frac{1}{\sqrt{2}}\left[|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\right]}_{\text{SINGOLETTO } (\chi_{ANTI})}$

- $\psi(1,2) = \underbrace{\frac{1}{\sqrt{2}}\left[\varphi_a(1)\,\varphi_b(2) - \varphi_a(2)\,\varphi_b(1)\right]}_{\varphi_A(1,2)} \cdot \underbrace{\begin{cases} |\uparrow\uparrow\rangle \\ \frac{1}{\sqrt{2}}\left[|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\right] \\ |\downarrow\downarrow\rangle \end{cases}}_{\text{TRIPLETTO } (\chi_{SIM})}$

① <u>STESSO STATO</u> $\boxed{a=b} \Rightarrow \varphi_a = 0 \Rightarrow \psi(1,2) = \varphi_S(1,2)\cdot\chi_{SINGOLETTO}$

$\Rightarrow$ Sullo STESSO STATO POSSO METTERE SOLO ELETTRONI CON <u>SPIN OPPOSTO</u>

(PRINCIPIO di esclus. di PAULI)

$(n, m_s) \rightarrow$  $n=1$ (almeno un num. quantico diverso)

② <u>DISTANZA MEDIA</u>

$\vec{r}_1 \longrightarrow \vec{r}_2 \Rightarrow \varphi_S = \sqrt{2}\,\varphi_a(2)\,\varphi_b(2) \Rightarrow |\varphi_S|^2 \rightarrow 2|\varphi_a(2)\,\varphi_b(2)|^2$

(Prob. di trovare 1 vicino a 2) ↳ SINGOLETTO

$\Rightarrow \varphi_A = 0 \rightarrow$ TRIPLETTO $\Rightarrow |\varphi_A|^2 = 0$

(la prob. di trovare due el. con lo stesso SPIN vicini $\rightarrow 0$)

$\Rightarrow \varphi_{Distinguibili} = \varphi_a(1)\,\varphi_b(2) \Rightarrow |\varphi_D|^2 \rightarrow |\varphi_a(2)\,\varphi_b(2)|^2$

↳ Non ho l'obbligo di simmetrizzazione

↳ MINORE RISPETTO AL CASO DI SINGOLETTO

$|\vec{r}_2 - \vec{r}_1|$

↳ Dist. tra : due fermioni

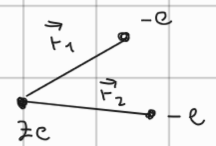
$$\langle |\vec{r}_2 - \vec{r}_1| \rangle = \int_{\mathbb{R}^3} |\vec{r}_2 - \vec{r}_1| \, |f|^2 \, d\vec{r}_1 \, d\vec{r}_2$$

$$\Rightarrow \boxed{\langle |\vec{r}_2 - \vec{r}_1| \rangle_S < \langle \vec{r}_2 - \vec{r}_1 \rangle_D < \langle |\vec{r}_2 - \vec{r}_1| \rangle_A}$$

(nello conf. di tripletto le part. sono mediam. più lontane che nello conf. di sing.)

ATOMO di ELIO:

$\ulcorner Z=2 \lrcorner$



$$\hat{H} = -\frac{\hbar^2}{2m}\Delta_1 - \frac{\hbar^2}{2m}\Delta_2 - \frac{Ze^2}{4\pi\varepsilon_0 r_1} - \frac{Ze^2}{4\pi\varepsilon_0 r_2} + \underbrace{\frac{e^2}{4\pi\varepsilon_0|\vec{r}_2-\vec{r}_1|}}_{V_{ee}}$$

SI RISOLVE CON METODI MATEMATICI

$$\psi = \varphi(\vec{r}_1, \vec{r}_2)\,\chi_{SPIN}$$

1. TRASCURO REPULSIONE V_{ee} → APPROSS. di part. INDIPENDENTI
2. " SPIN

$$\rightarrow \hat{H} = \hat{H}_1 + \hat{H}_2 \Rightarrow \varphi(\vec{r}_1, \vec{r}_2) = \varphi_{n_1 l_1 m_1}(\vec{r}_1)\,\varphi_{n_2 l_2 m_2}(\vec{r}_2)$$

$$\Rightarrow \hat{H}_1\,\varphi_{n_1 l_1 m_1}(\vec{r}_1) = E_1\,\varphi_{n_1 l_1 m_1}$$

CONOSCO GIÁ I RISULTATI

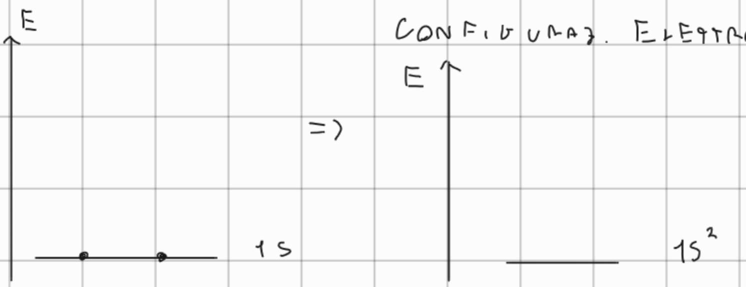
$$\hat{H}_2\,\varphi_{n_2 l_2 m_2}(\vec{r}_1) = E_2\,\varphi_{n_2 l_2 m_2}$$

$$E_1 = -\frac{Z^2 R_y}{n_1^2} \qquad E_2 = -\frac{Z^2 R_y}{n_2^2} \qquad \boxed{E = E_1 + E_2} \Rightarrow \text{STATO FONDAMENT.}$$

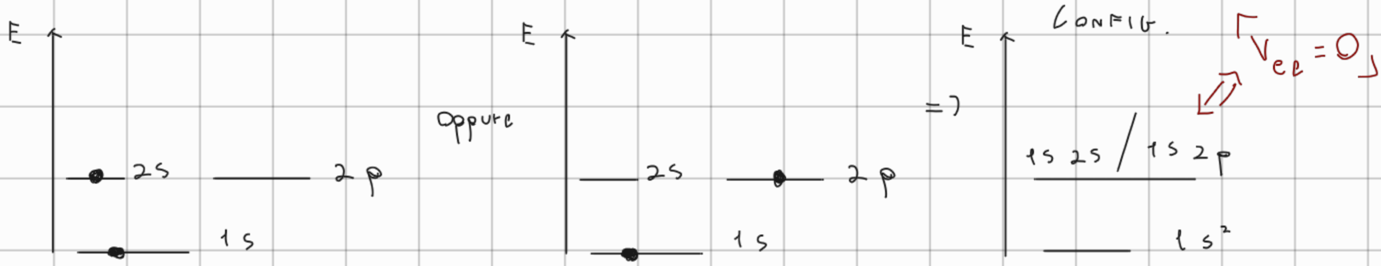
$$\Rightarrow E = -8\,R_y = -108.8\ eV$$

$$\varphi(\vec{r}_2, \vec{r}_1) = \varphi_{100}(\vec{r}_1)\,\varphi_{100}(\vec{r}_2)$$

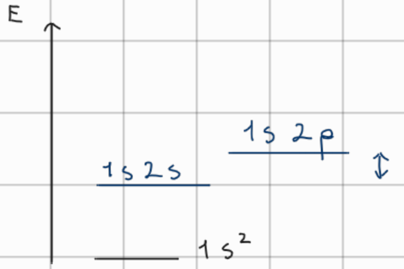
CONFIGURAZ. ELETTRONICA



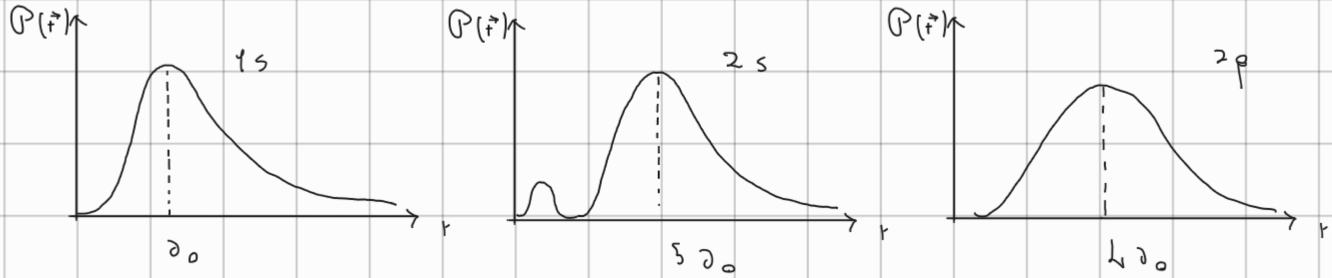
→ STATI ECCITATI:



$\underline{V_{ee} \neq 0:}$ $\Rightarrow$



$E_{SF} = -109\, eV \Rightarrow \boxed{E_{SF} \simeq -79\, eV}$



1s e 2p SI SOVRAPPONGONO DI PIÙ di 1s e 2s $\Rightarrow$ gli è di

1s e 2p si repellono maggiormente

STATO FOND. $\Rightarrow$ $1s^2$

1° STATO ECC. $\Rightarrow$ 1s 2s

2° STATO ECC. $\Rightarrow$ 1s 2p

$\longrightarrow$ AGGIUNGO LO <u>SPIN</u>:

Parte spaz.: $\psi(\vec{r}_1, \vec{r}_2) = \pm\, \psi(\vec{r}_2, \vec{r}_1)$

$$\psi_A = \frac{1}{\sqrt{2}}\left[\psi_{n_1 l_1 m_1}(\vec{r}_1)\, \psi_{n_2 l_2 m_2}(\vec{r}_2) - \psi_{n_1 l_1 m_1}(\vec{r}_2)\, \psi_{n_2 l_2 m_2}(\vec{r}_1)\right]$$

$\hookrightarrow$ <u>STATI ORTO</u>

$$\psi_S = \frac{1}{\sqrt{2}}\left[\psi_{n_1 l_1 m_1}(\vec{r}_1)\, \psi_{n_2 l_2 m_2}(\vec{r}_2) + \psi_{n_1 l_1 m_1}(\vec{r}_2)\, \psi_{n_2 l_2 m_2}(\vec{r}_1)\right]$$

$\hookrightarrow$ STATI PARA

- PARTE di SPIN:

$\chi_{SPIN}(1,2) = \longrightarrow \chi_{TRI}(1,2)$:

- $\longrightarrow \chi_+(1)\,\chi_+(2) \longrightarrow S_z = \hbar$
- $\longrightarrow \frac{1}{\sqrt{2}}\left[\chi_+(1)\,\chi_-(2) + \chi_+(2)\,\chi_-(1)\right] \rightarrow S_z = 0$
- $\longrightarrow \chi_-(1)\,\chi_-(2) \rightarrow S_z = -\hbar$

$\longrightarrow \chi_{SING}(1,2) = \frac{1}{\sqrt{2}}\left[\chi_+(1)\,\chi_-(2) - \chi_+(2)\,\chi_-(1)\right] \rightarrow S_z = 0$

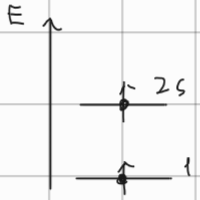
$\psi(1,2) \longrightarrow \varphi_A \, \chi_{TRI}$

$\quad\quad\quad\ \ \hookrightarrow \varphi_S \, \chi_{SING}$

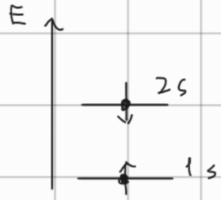
→ STATO FOND.:

$$\varphi(\vec{r}_1, \vec{r}_2) = \varphi_{100}(\vec{r}_1)\, \varphi_{100}(\vec{r}_2) \Rightarrow \boxed{\psi(1,2) = \varphi_{100}(\vec{r}_1)\, \varphi_{100}(\vec{r}_2)\, \chi_{SING}}$$

1° STATO ECC.:

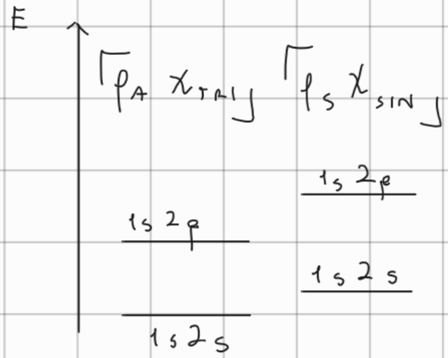


opp.



IDEM. per 2° STATO ECC.

$\langle |\vec{r}_2 - \vec{r}_1| \rangle_{TRI} > \langle |\vec{r}_2 - \vec{r}_1| \rangle_{SING}$ [se cons. V_{ee} χ_{TRI} è favorito in En. rispetto a χ_{SING}]

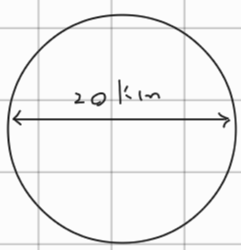


1° stato Ecc.: $\frac{1}{\sqrt{2}}\left[\varphi_{100}(\vec{r}_1)\,\varphi_{200}(\vec{r}_2) - \varphi_{100}(\vec{r}_2)\,\varphi_{200}(\vec{r}_1)\right]\chi_{TRI}$

2° stato Ecc.: $\frac{1}{\sqrt{2}}\left[\varphi_{100}(\vec{r}_1)\,\varphi_{21m_2}(\vec{r}_2) - \varphi_{100}(\vec{r}_2)\,\varphi_{21m_1}(\vec{r}_1)\right]\chi_{TRI}$

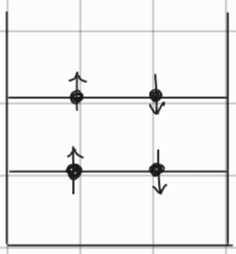
PRESSIONE di DEGENERAZIONE:

Stella di neutroni



$\rho \simeq 10^{17}\, kg/m^3$

i neut. sono ferm. e per Pauli: non posso stare tutti nello stesso stato quantico, quindi non implodono



SONO IN UNA BUCA di pot. a causa della gravità

quindi $E_K \uparrow$ → si chiama pressione di degenerazione che si oppone alla gravità